

Heena Khan

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EDUCATION

Stanford University

Expected Graduation: June 2027 (B.S.)

B.S. Computer Science (Artificial Intelligence) | Minor: Biology

Relevant Coursework: Operating Systems, Data Structures & Algorithms, Databases, Machine Learning, Cloud Computing, Probability & Statistics, Computer Vision, Compilers

EXPERIENCE

Software Engineering Intern — Data Infrastructure & Full Stack

Santa Clara, CA | June 2026 – October 2026

Roche / Genentech

- Built domain-specific AI agent translating natural language into SQL over a 25+ table production PostgreSQL database; designed and evaluated ground-truth Q&A pipeline to measure end-to-end agent accuracy — same pattern as LLM-over-data systems at AI platform companies.
- Engineered schema drift detection pipeline that ran before every ETL load, validating column types, value distributions, and sheet structure to prevent silent data corruption across production datasets.
- Built containerized Python/SQL ETL pipelines using Docker with automated CI/CD schema validation, ingesting and normalizing complex PCR and NGS assay records at scale on Linux cluster storage.
- Architected PostgreSQL schemas with multi-column indexing and partial indexes across 25+ tables; engineered full-stack Django and React application with real-time data visualizations shipped to production.

AI Research Engineering Intern

Palo Alto, CA | June 2025 – September 2025

Neurotrack Technologies

- Engineered high-throughput Python/SQL feature pipelines, ingesting and normalizing 10,000+ multimodal assessment records across 9 heterogeneous input schemas for downstream ML training.
- Designed automated data validation and noise-filtering algorithms to detect schema drift and artifact corruption, increasing dataset reliability for training pipelines.
- Trained and evaluated regression and curve-fitting models to map assessment trajectories, using statistical metrics to prune 6 redundant data variants.

Stanford Tech Fellowship

Stanford, CA | June 2023 – September 2024

Pull for Progress

- Engineered geospatial ML models in Python using multidimensional health metrics to predict risk profiles across 50,000+ data points; built reproducible processing pipelines with Pandas and NumPy.
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PROJECTS

AMANI — AI-Enabled Health Mapping System | Stanford Healthcare Design Challenge Finalist

Python, React Native, TensorFlow Lite, FastAPI, PostgreSQL, Computer Vision

- Designed end-to-end data pipeline for offline hazard report ingestion: local SQLite queue with geo-proximity deduplication, exponential backoff sync, and encrypted writes to PostgreSQL backend with full audit trail per record.
- Optimized on-device inference pipeline using INT8 quantized TFLite models for sub-200ms classification under RAM and compute constraints; built FastAPI microservice backend handling ingest and human-in-loop verification.

Khaista Boutique — Full-Stack E-Commerce Platform

TypeScript, React, Node.js, PostgreSQL, Drizzle ORM, Docker, GitHub Actions

- Built end-to-end platform with React + TypeScript frontend, Node.js/Express REST API, PostgreSQL with Drizzle ORM, Zod input validation, Stripe payments, and automated CI/CD via GitHub Actions.
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SKILLS

Languages: Python, Java, C++, SQL, TypeScript, JavaScript

Data & Infrastructure: PostgreSQL, Docker, Linux, Bash, Git, CI/CD, ETL Pipelines, Schema Drift Detection, REST APIs

AI / ML: LLMs, NL-to-SQL Agents, Prompt Engineering, Computer Vision, TensorFlow Lite, Statistical Modeling, Pandas, NumPy